WORKED SOLUTION

STEP I 1994, Question 9

Mechanics • Projectile Motion

STEP I 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q9

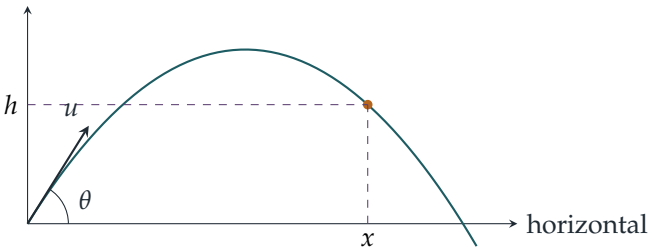
The key idea

The firing angle θ is the thing we don't know, so eliminate it: the horizontal and vertical equations each contain $u\cos\theta$ or $u\sin\theta$, and squaring and adding removes θ completely. The resulting equation is a quadratic in t^2, and a quadratic only has real roots when its discriminant is non-negative. That condition is exactly the bound on the range.

Solution

The equation in t

Model the cannon-ball as a particle moving under gravity alone, fired from the origin at angle θ above the horizontal. After time t it is at horizontal distance x and height h, so its distance from the cannon is $\sqrt{x^2+h^2}$, as required.



Using constant acceleration g downwards and no horizontal force,

$$x = ut\cos\theta,$$
$$h = ut\sin\theta - \tfrac{1}{2}gt^2.$$

Isolate the θ terms:

$$ut\cos\theta = x, \qquad ut\sin\theta = h + \tfrac{1}{2}gt^2.$$

Squaring and adding, and using $\cos^2\theta + \sin^2\theta = 1$,

$$\begin{aligned} u^2t^2 &= x^2 + (h + \tfrac{1}{2}gt^2)^2 \\ &= x^2 + h^2 + ght^2 + \tfrac{1}{4}g^2t^4, \end{aligned}$$

which rearranges to

$$\tfrac{1}{4}g^2t^4 - (u^2 - gh)t^2 + h^2 + x^2 = 0,$$

as required.

The bound on the range

Now fix u and h, and ask which horizontal distances x are possible at height h. If the ball is at (x, h) at some time t, then $T = t^2$ is a real root of

$$\tfrac{1}{4}g^2T^2 - (u^2 - gh)T + (h^2 + x^2) = 0.$$

Ask yourself The equation involves x, h, u and t, but we want a statement about x alone. What must be true of a quadratic if it has a real root?

A real root needs a non-negative discriminant:

$$\begin{aligned} (u^2 - gh)^2 - 4 \cdot \tfrac{1}{4}g^2(h^2 + x^2) &\geq 0 \\ u^4 - 2u^2gh + g^2h^2 - g^2h^2 - g^2x^2 &\geq 0 \\ g^2x^2 &\leq u^2(u^2 - 2gh). \end{aligned}$$

Since $u^2 > 2gh$ the right-hand side is positive, and taking non-negative square roots gives

$$x \leq \frac{u\sqrt{u^2 - 2gh}}{g}.$$

So the horizontal range at height h can never exceed this value, which is what we had to show. (If $u^2 < 2gh$ the inequality can't hold for any x: the height h is above the greatest height $u^2/2g$ the ball can reach.)

The bound is attained

Equality means the discriminant is zero, so there is a double root

$$t^2 = T = \frac{u^2 - gh}{2 \cdot \frac{1}{4}g^2} = \frac{2(u^2 - gh)}{g^2}.$$

This is positive (because $u^2 > 2gh \geq gh$ if $h \geq 0$, and trivially if $h < 0$), so it gives a genuine time $t > 0$. We still have to check that some angle θ produces this time with $x = u\sqrt{u^2 - 2gh}/g$. From the two component equations we need

$$\sin\theta = \frac{h + \frac{1}{2}gt^2}{ut}, \qquad \cos\theta = \frac{x}{ut}.$$

With the value of t^2 above, $h + \frac{1}{2}gt^2 = h + (u^2 - gh)/g = u^2/g$, so

$$\sin\theta = \frac{u}{gt}, \qquad \cos\theta = \frac{\sqrt{u^2 - 2gh}}{gt}.$$

Ask yourself We have chosen values for $\sin\theta$ and $\cos\theta$ separately. How do we know a single angle has both?

Both values are positive, and

$$\sin^2\theta + \cos^2\theta = \frac{u^2 + u^2 - 2gh}{g^2t^2} = \frac{2(u^2 - gh)}{g^2t^2} = 1$$

by the value of t^2. So there is exactly one angle θ between 0 and $\frac{1}{2}\pi$ that works, namely

$$\tan\theta = \frac{u}{\sqrt{u^2 - 2gh}}.$$

Firing at this angle, the ball is at height h at time t with horizontal distance exactly $u\sqrt{u^2 - 2gh}/g$, so the bound is always attained when $u^2 > 2gh$.

Notice that $\tan\theta > 1$ when $h > 0$: to reach furthest at a given height above the cannon, we fire more steeply than 45°. Also the vertical velocity at that moment is $u\sin\theta - gt = (u^2 - g^2t^2)/(gt) = (2gh - u^2)/(gt) < 0$, so the ball reaches its furthest point at height h on the way down.

Checks

- With $h = 0$ the bound is u^2/g and $\tan\theta = 1$: the familiar maximum range on level ground, at 45°.
- With $u = 20\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $h = 10\,\mathrm{m}$ the bound is about 29.15 m, at $\theta \approx 54.5°$.
- We confirmed numerically that the quartic holds along random trajectories, and that scanning thousands of firing angles never beats the bound but gets within 10^{-4} of it. The angle above hits it exactly.

Common slips

- Trying to find the best θ by differentiating x with respect to θ. You can, but it's messy. The discriminant does the optimisation for you.
- Forgetting the final step: a zero discriminant gives a value of t, but you must still show that a real angle produces it, by checking $\sin^2\theta + \cos^2\theta = 1$.

Beyond STEP

The points the ball can reach at speed u fill a region bounded by the *envelope* of all the trajectories, the "parabola of safety" $y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}$. Rearranging the bound here shows that (x, h) is reachable exactly when it lies on or under this parabola. Envelopes of families of curves come from the same "double root" idea, and they come up again in the calculus of variations and in optics (caustics). For more on projectiles and the mechanics behind them, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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